FINAL TECH STACK (with NiceGUI Only)

1. Programming Language

- **Python 3.10+**

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2. AI/ML Stack

- **scikit-learn** (Model Training & Inference)
- **pandas, numpy** (Data Handling)
- **SHAP** (Explainable AI)
- **imbalanced-learn** (Fraud class imbalance handling)
- **joblib** (Model export)

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3. Blockchain Layer (Custom)

- **Custom Blockchain in Python**
  - SHA-256 hashing (hashlib)
  - Digital Signatures (PyCryptodome)
  - Immutable Ledger stored in JSON / SQLite

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4. Backend Framework

- **FastAPI** (preferred for performance)
  or
- **Flask** (lightweight offline choice)

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5. Frontend / Dashboard (REQUIRED)

NiceGUI (FINAL CHOICE)

- Fully offline
- Cross-platform
- Mobile-responsive
- Beautiful UI

- Easy chart embedding
- Runs inside the same Python process
- Perfect for academic + enterprise dashboards

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6. Database

- **SQLite3**
- **SQLAlchemy ORM** (optional, for model abstraction)

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7. Visualization Libraries

- **Matplotlib**
- **Seaborn**
- **Plotly (optional)**
  All charts exported as PNG as per project requirement.

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8. Deployment

- **PyInstaller** (EXE/AppImage/.APP)
- **venv** (isolated environment)

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9. Security

- **PyCryptodome** (Digital signatures)
- **hashlib** (SHA-256 hashing)
- **Input validation (pydantic/fastapi)**

SHORT FINAL TECH STACK (FULLY FINAL)

Use this one in all documents:

Python + scikit-learn + SHAP + Custom Blockchain + SQLite + FastAPI + NiceGUI Dashboard + Matplotlib/Plotly + PyInstaller